

Isha Chaudhary

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Education

University of Illinois Urbana-Champaign

2022 - 2027

Ph.D. in Computer Science; Guide: Prof. Gagandeep Singh

University of Illinois Urbana-Champaign

2022 - 2025

M.S. in Computer Science with *Thesis*; Guide: Prof. Gagandeep Singh

GPA: 3.97/4.00

Indian Institute of Technology Delhi, India

2018 - 2022

B.Tech in Electrical Engineering (minor in CS); GPA: 9.3/10.0

Class Rank: 1/51

Research Interests

Safety and trustworthiness of frontier AI systems: certifying catastrophic risks and adversarial robustness with formal guarantees for LLM and agentic system behaviors. <https://certifyllm.com/>

Selected Publications

- Isha Chaudhary*, Vedaant Jain*, Kavya Sachdeva, Sayan Ranu, Gagandeep Singh. *Revealing Interpretable Failure Modes of VLMs*. Arxiv 2026. (* indicates equal contribution)
- Isha Chaudhary, Vedaant Jain, Gagandeep Singh. *QuaCer-C: Quantitative Certification of Knowledge Comprehension in LLMs*. **AISTATS 2026 Spotlight**, also in SeT LLM Workshop @ ICLR 2024.
- Chengxiao Wang, Isha Chaudhary, Qian Hu, Weitong Ruan, Rahul Gupta, Gagandeep Singh. *How Catastrophic is Your LLM? Certifying Risks in Conversation*. **ICLR 2026**.
Impact: Deployed in Amazon's Nova evaluation suite
- Isha Chaudhary, Qian Hu, Manoj Kumar, Morteza Ziyadi, Rahul Gupta, Gagandeep Singh. *Quantitative Certification of Bias in Large Language Models*. **ICLR 2025**.
- Isha Chaudhary, Vedaant Jain, Avaljot Singh, Kavya Sachdeva, Sayan Ranu, Gagandeep Singh. *Lumos: Let there be Language Model System Certification*. PAGe workshop @ PLDI 2026.
Impact: In deployment within National Research Platform's model cards
- Jehyeok Yeon, Isha Chaudhary, Gagandeep Singh. *Quantifying Distributional Robustness of Agentic Tool-Selection*. AIWILD @ ICLR 2026.
- Jason Vega*, Isha Chaudhary*, Changming Xu*, Gagandeep Singh. *Bypassing the Safety Training of Open-Source LLMs with Priming Attacks*. **ICLR 2024** (Tiny Papers track). (* indicates equal contribution)
Impact: Attack Success Rate $\uparrow 3.3\times$ over baselines on harmful behavior benchmarks
- Isha Chaudhary, Shuyi Lin, Cheng Tan, Gagandeep Singh. *Specification Generation for Neural Networks in Systems*. **ICSE 2026** (Poster), also in ML4Wireless Workshop @ ICML 2025.
- Isha Chaudhary, Alex Renda, Charith Mendis, Gagandeep Singh. *COMET: X86 Cost Model Explanation Framework*. **MLSys 2024**, also in XAI-in-Action Workshop @ NeurIPS 2023.

Experience

Amazon AGI

Boston, USA

Applied Scientist Intern with Dr. Rahul Gupta and Dr. Weitong Ruan

May 2026 - Aug 2026

Amazon Web Services

Seattle, USA

Applied Scientist Intern with Dr. Ganyu Teng and Dr. Shawn Zhang

Sep 2025 - Dec 2025

- Developed a principled framework to stress-test scalable-oversight-enabled agentic systems via adversarial prompt injections, using CVaR to expose worst-case alignment failures hidden by average-case metrics
- Finetuned Qwen-1.5B as a transferable adversary with GRPO to generate effective prompt-injection attacks against scalable-oversight judge models.

Microsoft Research

Redmond, USA

Research Intern with Dr. Ryan Beckett, Dr. Siva Kakarla, Prof. Francis Yan, Dr. Behnaz Arzani

May 2025 - Aug 2025

Developed LLM-powered semantic analysis of system logs for anomaly detection, leveraging frontier model capabilities for safety-critical infrastructure monitoring.

Microsoft AI (Bing)

Redmond, USA

Data Science Intern

May 2024 - Aug 2024

Developed machine learning models for revenue optimization in Bing search.

Adobe Research (Cloud Tech group)

Bengaluru, India

Research Intern / Guide: Dr. Gaurav Sinha / **US patent no. US 2023/0274310 A1**

May 2021 - Aug 2021

Multi-task Learning with Label Proportions: Jointly disaggregated multiple aggregate labels with individual features

Blockchain IoT Security Project: DST-JST Grant on IoT Security

IIT Delhi, India

Research Assistant / Guide: Prof. Subodh Sharma, IIT Delhi / Denso Intl. India

Sept 2020 – May 2021

Designed & developed Hyperledger Fabric-based supply chain system for real-time order traceability & security

Awards

- Awarded **Best Presentation Award** (Security and Privacy track) at CSL student conference 2025 (2025)
- Awarded **Institute Silver Medal & Prof. C.S. Jha Memorial Excellence Award** at IIT Delhi (2022)
- **Top 7% merit award**: In top 7% high GPA students at IIT Delhi for 5 semesters: semesters 3, 4, 6, 7, 8 (2019-22)
- **Design & Innovation Summer Award (MHRD Grant, Govt. of India)** (2019)
- **KVPY**: Qualified for the Kishore Vaigyanik Protsahan Yojana Fellowship (IISc); All India Rank 232 (2016)
- **NTSE**: Received National Talent Search Scholarship, given to top 1000 students nationwide by NCERT (2015)

Talks

Certifying LLMs

- NSF's National AI Research Resource (NAIRR) Pilot 2026
- Microsoft Research Redmond and Microsoft Research India 2025
- CSL student conference 2025
- IIT Delhi 2025
- Amazon-Illinois Center on AI for Interactive Conversational Experiences 2024

Teaching Smarter with AI: How to Use AI in Classrooms Without Losing Control

- Graduate TA training academy organized by CITL, UIUC 2025

Exploring COMET: Neural Cost Model Explanation Framework

- Google Compiler Research reading group 2024
- VMWare Research 2024

Teaching

CS 477: Formal Software Development Methods

UIUC, USA

Guest Lectures on formally certifying frontier models

Apr 2026

Grad Academy, CITL

UIUC, USA

Instructor: Isha Chaudhary, UIUC

Aug 19, 2025

Advanced Topics in Programming Systems - Trustworthy AI Systems (CS521)

UIUC, USA

Instructor: Prof. Gagandeep Singh, UIUC

Aug 2024 – Dec 2024

Calculus (MTL100)

IIT Delhi, India

Instructor: Prof. Sivananthan Sampath, IIT Delhi

Aug 2021 – Dec 2021

Grants

- Awarded **ICML 2023 Travel Grant** (2023)
- Awarded scholarship for **Programming Languages Mentoring Workshop @ PLDI 2023** (2023)
- **NSF Travel Grant for IEEE SaTML 2023** (2022)

Service

- Area Chair for MLSys 2025.
 - Reviewer for ICML 2026; KDD2026; COLM 2026; ICLR 2025, 2026; FAccT 2025, 2026; MLSys 2025; AIES 2024, 2025; Tiny Papers @ ICLR 2024.
 - Artifact Evaluator for VMCAI 2024.
 - Main organizer for Formal Methods and Software Engineering seminar, Spring 2024 & Fall 2024, at UIUC.
 - Mentor for the Undergraduate Research Apprenticeship Program (URAP) at UIUC, 2024-25
 - Mentor in the Undergraduate Research in Scientific Advancement (URSA) program at UIUC, 2024-25
 - Mentor in SIGPLAN-M long-term mentorship programme
 - Mentor in K-12 Broadening Participation in Computing program at UIUC, Summer 2023.
- (Guided a team of high-school students in successfully learning & conducting research on LLMs.)

Undergraduate Thesis Project

Security & Privacy of SCADA systems & Microgrids

IIT Delhi, India

Guides: Prof. Anupam Joshi & Prof. B.K. Panigrahi

Aug 2021 - Dec 2021

- Developed Multi-task data-driven real-time Intrusion detection system for digital twin of AC Microgrid (AUC: 0.85)